

Practice Problems

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Problem 1: Algorithm Properties & correctness

a) Input

Both take an array temperature[]
an integer n.

Input property satisfied ✓

3. Definiteness

Each step in both algos
are well defined.

2. Output

Both return average value of the
temperature.

Output property satisfied ✓

4. Finiteness

Both loops are finite

5. Effectiveness

Both are effective but algo B is not correct. The division
operation has been used wrongfully.

b) Step by step execution

Temp = [20, 25, 22, 24, 21]

n = 5

Algo A
index

i = 1

i = 2

i = 3

i = 4

i = 5

Operation

0 + 20

20 + 25

45 + 22

67 + 24

91 + 21

Total

20

45

67

91

112

Average = Total / n

= 112 / 5

= 22.4

Algo B

sum = temp[1] = 20

first loop

	sum
25	45
22	87
24	91
21	112

second loop

for 1 to 5

sum = sum/n;

1	3
1	22.4
2	4.48
3	0.896
4	0.1792
5	0.03584

c) Logical error in Algo B

Mathematically — sum/n^n instead of $\frac{\text{sum}}{n}$

e) test case

1. works for array
2. works for a single element
3. negative num
4. same val
5. Large
6. mixed value

A	B
✓	✓
✓	✓
0	Error
10, 10, 10	10
array	single sum
Right Ans.	wrong Ans

2.

a) Algo A

worst case =
comparisons = n

Algo C

hash table

1 comparison

Algo B

worst case $\log_2 n$

b) Linear
Binary
Hash Table

Time complexity
 $O(n)$
 $O(\log n)$
 $O(1)$

Hash-table
checks every element
search space halves
direct key access

c) n

100

1000

10000

100,000

1,000,000

10,000,000

Linear search

100

1000

10000

100000

1000000

10000000

Binary search

7

10

14

17

20

24

Hash table

1

1

1

1

1

1

d) 50,000 searches

Linear search

cannot handle
~~it~~

Binary search

can handle

Hash Table

Best performance

e) $k \cdot n > n \log n + k \log n$

$k > \log n$

$\log n = 24$

Q3

a) $T(n, m) = m \times n$

Nested loops — 1

Hash set —
 $O(n + m)$

b) space complexity

nested loop (A) —

$O(1)$

hash set (B) —

$O(n)$

c) given value.

$n = 40$

$m = 5$

students per day = 100,000

Implementation A

$$m \times n = 5 \times 40$$

$$= 200$$

Total

$$200 \times 100,000$$

$$\rightarrow 20,000,000$$

Implementation B

$$40 \times 8 = 320$$

$$320 \times 100,000$$

$$= 32,000,000$$

d) Implementation B is better

e) $\rightarrow$ m is very small $\rightarrow$ nested loop
 m is big $\rightarrow$ Hash set

Problem 4

a) Bubble sort

Worst case $\rightarrow T(n) = n^2$

Merge sort

$$T(n) = n \log n$$

b) Merge sort $\rightarrow \Theta(n \log n)$ = Exact tight bound (upper & lower)
 Quick sort $\rightarrow O(n \log n)$ = upper bound only

c) Data type

Nearly sorted

Random

Reverse sorted

Best Algo

Insertion sort

Merge sort / Quick sort

Merge sort

Justification

$O(n)$ - best case

$O(n \log n)$ - avg

$\Theta(n \log n)$ - Guarantee

d) $n < 50$

~~Insertion~~
 Insertion sort

$(50 \leq n < 10,000)$

Quick sort

$(n \geq 10,000)$
 Merge sort

e) Merge sort - small input
 Merge sort - Realistic data sizes